VSAM Key and Duplicate-Key Tests

COBOL Program Report

1. Aim

To simulate VSAM key validation by inserting account IDs, detecting duplicate keys, and testing batch rollback. The program displays the result of each operation and the final record count. Account IDs are stored as alphanumeric values so leading zeros are preserved.

2. Algorithm

1. Initialise the account table with account IDs 00001, 00002 and 00003.
2. Display the initial account record count.
3. To insert an account, set the duplicate flag to N and search all existing account IDs.
4. If the new account ID already exists, set the duplicate flag to Y and display DUPLICATE KEY.
5. If the key is new and the table has space, increase the record count, store the key and display INSERT SUCCESS.
6. For the batch test, copy the current table and record count as a backup.
7. Insert account 00005, then attempt to insert account 00003, which already exists.
8. If the duplicate flag is Y, restore the saved table and count and display BATCH ROLLBACK SUCCESS; otherwise display BATCH COMMITTED.
9. Display the final record count.

3. COBOL Code

```
       IDENTIFICATION DIVISION.
       PROGRAM-ID. VSAM-KEY-TEST.

       DATA DIVISION.
       WORKING-STORAGE SECTION.

       01 ACCOUNT-TABLE.
          05 ACCOUNT-ID OCCURS 10 TIMES PIC X(5).

       01 ACCOUNT-COUNT PIC 99 VALUE 3.
       01 I PIC 99.
       01 J PIC 99.
       01 FOUND-FLAG PIC X VALUE 'N'.

       01 NEW-ACCOUNT PIC X(5).

       01 BACKUP-TABLE.
          05 BACKUP-ID OCCURS 10 TIMES PIC X(5).

       01 BACKUP-COUNT PIC 99.

       PROCEDURE DIVISION.

           MOVE '00001' TO ACCOUNT-ID(1)
           MOVE '00002' TO ACCOUNT-ID(2)
           MOVE '00003' TO ACCOUNT-ID(3)

           DISPLAY 'INITIAL RECORD COUNT: '
                   ACCOUNT-COUNT

           DISPLAY ' '
           DISPLAY 'TEST 1: INSERT NEW ACCOUNT'

           MOVE '00004' TO NEW-ACCOUNT
           PERFORM INSERT-ACCOUNT

           DISPLAY ' '
           DISPLAY 'TEST 2: INSERT DUPLICATE KEY'

           MOVE '00002' TO NEW-ACCOUNT
           PERFORM INSERT-ACCOUNT

           DISPLAY ' '
           DISPLAY 'TEST 3: BATCH ROLLBACK'

           PERFORM BATCH-TEST

           DISPLAY ' '
           DISPLAY 'FINAL RECORD COUNT: '
                   ACCOUNT-COUNT

           STOP RUN.

       INSERT-ACCOUNT.

           MOVE 'N' TO FOUND-FLAG

           PERFORM VARYING I FROM 1 BY 1
               UNTIL I > ACCOUNT-COUNT

               IF ACCOUNT-ID(I) = NEW-ACCOUNT
                   MOVE 'Y' TO FOUND-FLAG
               END-IF

           END-PERFORM

           IF FOUND-FLAG = 'Y'
               DISPLAY 'DUPLICATE KEY: '
                       NEW-ACCOUNT
           ELSE
               IF ACCOUNT-COUNT < 10
                   ADD 1 TO ACCOUNT-COUNT

                   MOVE NEW-ACCOUNT
                       TO ACCOUNT-ID(ACCOUNT-COUNT)

                   DISPLAY 'INSERT SUCCESS: '
                           NEW-ACCOUNT
               ELSE
                   DISPLAY 'TABLE FULL'
               END-IF
           END-IF.

       BATCH-TEST.

           MOVE ACCOUNT-TABLE TO BACKUP-TABLE
```

```
            MOVE ACCOUNT-COUNT TO BACKUP-COUNT

            MOVE '00005' TO NEW-ACCOUNT
            PERFORM INSERT-ACCOUNT

            MOVE '00003' TO NEW-ACCOUNT
            PERFORM INSERT-ACCOUNT

            IF FOUND-FLAG = 'Y'
                MOVE BACKUP-TABLE TO ACCOUNT-TABLE
                MOVE BACKUP-COUNT TO ACCOUNT-COUNT

                DISPLAY 'BATCH ROLLBACK SUCCESS'
            ELSE
                DISPLAY 'BATCH COMMITTED'
            END-IF.
```

4. Expected Output

```
INITIAL RECORD COUNT: 03

TEST 1: INSERT NEW ACCOUNT
INSERT SUCCESS: 00004

TEST 2: INSERT DUPLICATE KEY
DUPLICATE KEY: 00002

TEST 3: BATCH ROLLBACK
INSERT SUCCESS: 00005
DUPLICATE KEY: 00003
BATCH ROLLBACK SUCCESS

FINAL RECORD COUNT: 04
```

5. Result

The program demonstrates successful insertion of a new account, rejection of a duplicate account ID, and rollback of a simulated batch after a duplicate-key failure. The insertion of 00005 is undone when the duplicate 00003 is detected. The final record count is 04: the original three records plus account 00004. This is a logic-level in-memory simulation and does not perform transactions against a real VSAM dataset.